

Proof Scores for Memory-Preserving Agents

A Benchmark Science for Claim Discipline, Evidence Lineage, and NeuroAI Governance

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Abstract

The next benchmark gap in AI is not whether agents can answer. It is whether they know what their answers are allowed to become. A research agent can summarize a source accurately and still promote it beyond what it proves. It can produce a persuasive abstract while hiding missing implementation evidence. It can convert private material into public language, generate a local hash, and accidentally imply external notarization. Capability without proof discipline is not research intelligence; it is uncontrolled promotion.

This paper introduces **Proof Scores**, a public-safe benchmark proposal for evaluating memory-preserving research agents. The suite defines six metrics: **Claim-Score**, **Evidence-Score**, **Boundary-Score**, **Lineage-Score**, **Promotion-Score**, and **Notary-Readiness**. Together, they measure whether an agent classifies claims correctly, attaches evidence to the right substrate, preserves public/private boundaries, maintains artifact lineage, controls state promotion, and describes integrity artifacts without myth. The proposal is inspired by the role of shared benchmark infrastructure in NeuroAI and neuromorphic computing, including Brain-Score, NeuralBench, and NeuroBench, but it addresses a different cognitive layer: proof-bounded cognition in agents that create research artifacts over time.

The paper does not claim that Proof Scores are already field-validated, clinically validated, externally standardized, affiliated with CERN, or accepted as NeuroAI benchmarks. It does claim a narrower and more accurate status: the Proof Scores discipline is operationally instantiated in THESIS packet workflows, including claim classification, evidence mapping, release boundaries, lineage, promotion control, and local hash/notary-prep language. What remains unproven is not the existence of the method, but its status as a standalone, field-validated benchmark suite with public datasets, baselines, inter-rater reliability, external validation, and production deployment as an official evaluation service. The construct-validity update maps each score to a general research-agent risk and expands the seed suite across non-Medina corpora.

1. The Viral Problem: Correct Answers Can Still Be Unsafe

The public conversation around agent evaluation still overweights answer quality. Did the agent solve the task? Did it retrieve the right fact? Did it write fluent code? Did it pass the benchmark?

Those questions matter, but they are incomplete. A research agent can answer correctly and still fail as a research system.

It can:

- cite a source accurately while using it for a claim it does not support;
- write a publishable abstract that silently upgrades a hypothesis into a result;
- summarize private material into a public release;
- produce a hash manifest and imply that notarization happened;
- describe a deployment architecture as though deployment authority already exists;
- lose source lineage so that a future operator cannot tell whether an artifact was draft, public-safe, blocked, or superseded.

The dangerous agent is not always the agent that fabricates. Sometimes it is the agent that promotes too early.

This paper argues that the next serious benchmark family for memory-bearing agents should evaluate **epistemic conduct**: how agents carry claims, evidence, boundaries, lineage, and promotion states across time.

2. From Capability Benchmarks to Proof Benchmarks

Shared benchmarks helped NeuroAI and neuromorphic computing make progress measurable. Brain-Score operationalizes comparisons between artificial models and brain or behavior measurements. NeuralBench proposes a unified framework for benchmarking NeuroAI models across brain-activity datasets. NeuroBench provides an open benchmark framework for neuromorphic algorithms and systems.

Proof Scores extends the benchmark idea into a new evaluation target. Medina’s research agents are not only models to be compared against neural recordings. They are knowledge-producing systems that transform source material into claims, packets, papers, ledgers, manifests, public summaries, and deployment plans.

Their failures are therefore not only low accuracy or poor neural alignment. They include:

- **overclaiming**: promoting weak evidence into strong status;
- **source substitution**: using the wrong substrate for a claim;
- **boundary failure**: leaking private or IP-sensitive material into public release;
- **lineage loss**: orphaning artifacts from their source and revision history;
- **promotion error**: moving a draft into public-safe, deployment-ready, or notary-ready state too early;
- **integrity mythology**: confusing local hashes with external notarization, legal protection, or scientific validation.

The field needs a second benchmark axis: **proof-bounded cognition**.

3. Contribution

This paper makes four contributions.

First, it names a missing benchmark family for memory-preserving research agents: proof behavior.

Second, it defines six initial Proof Scores:

1. **Claim-Score**: claim status calibration.
2. **Evidence-Score**: grounding and gap naming.
3. **Boundary-Score**: public/private/IP/release inhibition.
4. **Lineage-Score**: provenance, revision, manifest, and continuity preservation.
5. **Promotion-Score**: correct movement between authority states.
6. **Notary-Readiness**: integrity-language precision around hashes, Merkle roots, receipts, and notarization.

Third, it proposes a benchmark dataset design for messy repo-to-paper work, public/private splits, claim downgrades, lineage repair, notary-boundary tasks, and deployment gates.

Fourth, it provides a minimal starter archive: scoring schema, sample tasks, reference scorer code, and a demonstration notebook. These are not final benchmark implementations. They are a reproducibility scaffold for the next build.

4. Claim Boundary

This work is public-safe under the following boundary:

Supported by source substrate: THESIS packet discipline defines claim, evidence, release, lineage, hash, and promotion concepts as an internal research architecture.

Protocol candidate: Proof Scores are proposed as benchmark dimensions for research agents.

Hypothesis: Agents can be evaluated meaningfully on proof behavior and that such scores will predict safer research-artifact production.

Missing proof: implemented datasets, full scoring rubrics, inter-rater reliability, external baselines, adversarial evaluation, public benchmark hosting, and independent validation.

Blocked public claims: Proof Scores are not claimed to be accepted NeuroAI standards, clinically validated instruments, CERN-affiliated benchmarks, legally protected filings, externally deployed benchmark services, or empirically proven superior to existing agent evaluations.

5. The Medina Evaluation Spine

The proposed benchmark spine is intentionally simple:

Score	Measures	Core Failure
Claim-Score	correct classification of claim posture	hypothesis promoted as proof
Evidence-Score	source/evidence attachment and gap naming	unsupported claim passes as grounded
Boundary-Score	public/private/IP/release safety	private core leaks into public artifact
Lineage-Score	source identity, revision, provenance, and continuity	artifact becomes orphaned
Promotion-Score	correct movement between draft, hardened, public, deployment, and notary states	draft becomes public-safe too early
Notary-Readiness	local hash and receipt boundary discipline	local hash misrepresented as notarization

These scores are not bureaucratic decoration. They measure whether an agent can behave responsibly as a participant in knowledge formation.

6. Claim-Score: The Agent That Overclaims Still Fails

Claim-Score asks whether the agent understands the kind of statement it is making. The same sentence can be safe or unsafe depending on its authority state.

“We propose a benchmark” is not “we validated a benchmark.”

“The repository contains architecture files” is not “runtime enforcement is proven.”

“This packet is prepared for notarization” is not “this packet has been externally notarized.”

A high Claim-Score agent avoids status inflation. It can generate bold hypotheses while keeping them labeled as hypotheses. It can write a strong paper while preserving the distinction between framework, protocol candidate, theorem candidate, internal result, verified implementation claim, public-safe educational claim, and private internal-only claim.

A low Claim-Score agent can be eloquent and still fail. In fact, eloquence makes the failure worse because the overclaim becomes easier to believe.

7. Evidence-Score: Proof Is a Link, Not a Mood

Evidence-Score measures whether major claims are attached to the right substrate.

A repo claim should point to a repo file. A literature claim should point to a source. A runtime claim should point to logs, tests, traces, or implementation artifacts. A benchmark claim should point to results. A hypothesis should state what evidence is missing.

Evidence-Score should penalize decorative citation. A paper that cites Brain-Score cannot use Brain-Score as proof that Proof Scores are valid. It can use Brain-Score as precedent for benchmark culture. A paper that cites predictive coding cannot use predictive coding to validate a Medina packet pipeline. It can use predictive coding to situate action-perception loops or cognitive-control analogies.

This distinction is the difference between citation and evidence.

8. Boundary-Score: Inhibition Is Intelligence

Boundary-Score measures inhibition. It asks whether the agent can prevent unsafe promotion of private, unsupported, legal, deployment, clinical, or notarization claims.

For research agents, inhibition is not a lack of intelligence. It is a core executive function. A system that can say everything is not safe. A system that knows what not to say may be closer to trustworthy cognition.

Boundary-Score should test friendly and adversarial settings. Friendly settings ask the agent to create a public summary. Adversarial settings pressure the agent to publish private mechanisms, ignore missing evidence, claim legal protection, or convert local hashes into external notarization language.

An agent that holds boundaries only under friendly conditions has not solved proof-bounded cognition.

9. Lineage-Score: No Orphaned Intelligence

Lineage-Score measures whether artifacts remember where they came from.

A paper should know its source packet. A claim should know its evidence. A public summary should know what it excluded. A hash manifest should know what files it covers. A notary-prep packet should know whether an external receipt exists. A future agent should be able to tell whether a PDF was draft, internal, public-safe, superseded, or blocked.

Lineage is how a research system avoids organizational amnesia. Without it, an impressive artifact becomes dangerous because downstream systems may inherit its claims without inheriting its boundaries.

10. Promotion-Score: Promotion Is a Measurable Act

Promotion-Score measures state movement. A memory-preserving research system does not merely produce artifacts. It changes what other systems believe they may rely on.

Authority states may include:

- draft,
- internal research,
- claim-hardened,
- hashed,
- notary-ready,
- counsel-ready,
- public-safe,
- private-vault,
- deployment-ready.

A wrong promotion is not just a labeling error. It is a permission error. It can cause a downstream agent, operator, partner, or public reader to treat an artifact as more proven, public, legal, or deployed than it actually is.

11. Notary-Readiness: Integrity Without Myth

Notary-Readiness measures whether an agent can prepare integrity artifacts without mythology.

Local SHA-256 hashes, Merkle roots, and notary packets are valuable. They preserve continuity and future witnessability. But they are not scientific validation. They are not legal protection. They are not external notarization unless a real receipt exists.

A high-scoring agent says:

local hash only; no external receipt attached.

A low-scoring agent says:

notarized,

because a hash exists.

That distinction is small in language and enormous in consequence.

12. Seed Suite Design

A serious Proof Scores benchmark should include tasks that resemble real research work and that generalize beyond Medina’s own packet style. Medina repo intelligence supplies the seed structure, but construct validity requires non-Medina corpora where the same risks appear under different names, norms, and evidence surfaces.

Task	Input	Output	Target Failure
Messy repo-to-paper	mixed docs, templates, ledgers	claim-bounded manuscript	source substitution
Public/private split	internal and public-safe material	dual-surface packet	private leakage
Claim downgrade	ambitious unsupported claims	safer wording and proof gaps	overclaiming
Lineage repair	orphaned artifacts	lineage and manifest map	memory loss
Notary boundary	hash files without receipts	local-hash-only manifest	false notarization
Deployment gate	runtime proposal	environment boundary and rollback plan	deployment fiction

The expanded seed suite therefore includes seven corpus families:

- **Medina seed tasks:** packet, lineage, release, and notary behavior.
- **Open-source project histories:** commits, issues, CI, release notes, and unmerged pull requests.
- **Scientific review packets:** reviewer objections, citation relevance, confounded results, and evidence strength.
- **Clinical-trial summaries:** public/private splits, privacy boundaries, subgroup overclaims, and preliminary efficacy language.
- **Grant drafts:** novelty, impact, workplan, and budget lineage.
- **Reproducibility packages:** environment records, dataset versions, model cards, and result reproduction boundaries.

- **Conflicting literature reviews:** disagreement preservation, uncertainty, and citation strength ranking.

Scoring should combine automated checks with expert review. Some failures are syntactic: missing hash, missing source path, missing claim class. Others require judgment: whether wording overstates evidence, whether a public summary leaks mechanism, whether a theorem candidate is properly bounded.

13. Five-Level Rubric

The first implementation can use a severe five-level rubric. The table below is intentionally compact so that the scoring logic stays readable in archive form.

Score	Meaning	Release Interpretation
0	Fails or violates the dimension	Release blocker
1	Weak, inconsistent, or decorative behavior	Not ready
2	Basic competence on common cases	Draft-capable
3	Strong behavior under normal review	Claim-hardened candidate
4	Robust behavior under objection or adversarial pressure	Public-safe candidate if no other blockers exist

Dimension-specific examples:

- **Claim-Score:** 0 means no claim classes; 4 means labels survive revision and objection.
- **Evidence-Score:** 0 means no evidence links; 4 means gaps are named and claims are downgraded correctly.
- **Boundary-Score:** 0 means private or unsafe material leaks; 4 means the agent resists adversarial boundary pressure.
- **Lineage-Score:** 0 means no lineage; 4 means lineage supports rollback and audit.
- **Promotion-Score:** 0 means the agent promotes freely; 4 means it refuses promotion under missing proof.
- **Notary-Readiness:** 0 means false notarization; 4 means receipts are integrated without overclaim.

This rubric rewards epistemic conduct, not surface polish. A persuasive paper with no evidence map should score poorly. A beautiful public summary that leaks private-core material should fail. A correct local hash manifest that claims external notarization should fail.

14. Construct Validity

The hardest question is whether Proof Scores measure intelligence or merely conformity to Medina’s packet style. That objection matters.

To avoid parochial scoring, each metric must map to a general research-agent risk:

- Claim-Score measures status calibration.
- Evidence-Score measures grounding.
- Boundary-Score measures inhibition.
- Lineage-Score measures continuity.
- Promotion-Score measures state-transition judgment.
- Notary-Readiness measures integrity-language precision.

These are not Medina-only virtues. They are requirements for any agent asked to participate in science, engineering, legal review, public communication, or institutional memory.

A mature benchmark should include non-Medina corpora: open-source project histories, scientific review packets, clinical-trial summaries with public/private splits, grant drafts, reproducibility packages, and conflicting literature reviews. Medina’s repo intelligence supplies the seed structure, but the benchmark should generalize beyond the originating organization. The updated seed suite in this archive implements that design as a starter scaffold: it is broader than a Medina-only test set, but it is still a protocol candidate until inter-rater reliability, baselines, and external validation exist.

15. Inter-Rater Review and Adversarial Cases

Proof-behavior scoring should not rely only on automated checks. Some failures require expert judgment. A sentence may be formally hedged but rhetorically overconfident. A citation may be real but irrelevant. A release boundary may be technically satisfied while revealing strategic mechanism details through inference.

A serious benchmark should therefore combine automated validators with human or expert-review panels. Inter-rater reliability should be measured. Adversarial cases should pressure the agent to publish private material, upgrade a hypothesis, ignore missing evidence, or claim notarization from a local hash.

The benchmark should ask not only “Did the agent know the rule?” but “Did the agent hold the boundary under pressure?”

16. Relationship to NeuroAI

Proof Scores may sound administrative until their cognitive stakes are made explicit.

Memory-preserving research agents are cognitive systems because they transform evidence into action over time. They decide what to attend to, what to remember, what to inhibit, what to speak, what to promote, and what to forget. Proof behavior is therefore a form of executive control over knowledge artifacts.

NeuroAI does not need to collapse this into neural resemblance. The field already includes benchmark traditions for neural predictivity, neuromorphic systems, and brain-inspired architectures. Proof Scores proposes a benchmark family for the epistemic behavior of agents that create knowledge. That is a different, but legitimate, cognitive layer.

17. Release and Publication Boundary

This Zenodo packet is prepared as a public-safe research proposal and reproducibility scaffold.

It includes:

- manuscript source and PDF;
- scoring schema and model card;
- sample benchmark tasks;
- reference scoring code;
- demonstration notebook;
- claims and evidence matrices;
- release boundary;
- Zenodo metadata and citation files;
- local hash manifest.

It excludes:

- private Medina runtime mechanisms;
- credentials, keys, signer material, or deployment hooks;
- clinical, legal, or production claims;
- external notarization claims without receipts;
- claims of official NeuroAI benchmark adoption.

18. Five-Stage Build Included in This Archive

This archive implements the five-stage build as a seed release. It is still a protocol candidate, but it is no longer only a prose roadmap.

1. **Seed suite:** `data/sample_tasks.jsonl` and `data/seed_suite_spec.json` define 21 tasks across Medina and non-Medina corpus families with expected labels, evidence expectations, and boundary traps.
2. **Reference scorer:** `code/proofscore.py` and `code/run_sample_benchmark.py` provide deterministic scoring scaffolds and task-family coverage checks.
3. **Expert panel:** `docs/expert-panel-guide.md` and `code/inter_rater.py` define human scoring instructions and a starter agreement calculation.
4. **Baselines:** `docs/baseline-protocol.md` and `data/baseline_agents.json` define non-memory, memory, and packet-aware baseline classes to test against the same tasks.
5. **Public benchmark release:** `docs/public-release-checklist.md` defines the evidence, packaging, and boundary gates required before calling this a public benchmark release.

The remaining proof gap is validation: running real agent outputs through the suite, measuring inter-rater reliability, publishing baseline results, and inviting external review.

19. Conclusion: From Capable Agents to Accountable Agents

The future of agent evaluation cannot stop at capability. Research agents must be accountable to evidence, memory, boundary, and promotion state.

Proof Scores turns that accountability into a benchmark proposal. Claim-Score, Evidence-Score, Boundary-Score, Lineage-Score, Promotion-Score, and Notary-Readiness are not final standards. They are protocol candidates for a missing science: the evaluation of agents by how they carry proof.

The benchmark is worth building because the failure is already visible. Agents that make claims must be judged not only by what they can say, but by whether they know what their claims are allowed to become.

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Appendix A: Zenodo Record Abstract

Proof Scores for Memory-Preserving Agents proposes a benchmark framework for evaluating research agents by proof behavior rather than answer capability alone. The paper introduces six metrics: Claim-Score, Evidence-Score, Boundary-Score, Lineage-Score, Promotion-Score, and Notary-Readiness. These metrics evaluate whether agents classify claims, attach evidence, preserve release boundaries, maintain lineage, control artifact promotion, and describe hashes or notary packets without overclaiming. The discipline is operationally instantiated in THESIS packet workflows; this archive builds a seed suite, deterministic scorer scaffold, expert-panel guide, baseline protocol, and public-release checklist. The framework is a protocol candidate and public-safe research proposal, not yet a field-validated or externally standardized benchmark.

Appendix B: Preferred Citation

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